A Cohorts and analysis sequence

Discovery

1,350 questions
21,600 predictions

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Technical pilots

128 primary + 64 curated
Excluded from main estimates

→

Main evaluation

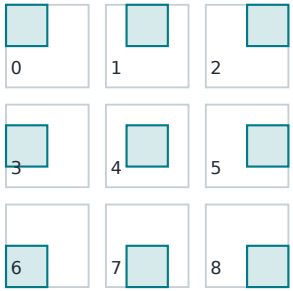
2,048 primary + 825 curated
Same questions, 2B and 4B

Frozen models and protocols; linkage excludes verified prior-cohort and cross-cohort reuse.

B One crop response per arm; matched visual budget

Nine overlapping windows

Width = height = 1/2 of image



Offsets: 0, 1/4, 1/2 per axis

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SigLIP 2 ranking

Question-to-crop similarity; no labels

Baseline response B

Question + global image

Guided response G

Global image + highest-ranked crop

Control response U_s

Global image + one of eight other crops
Seeds 1, 7, 42; repetitions retained

Guided / controls: same question, options, global image, extra-view dimensions and tokens.

C Separate available answers from improved decisions

Prespecified primary: 2B, primary cohort

$$D_i = \max(B_i, G_i) - \frac{1}{3} \sum_{s \in \{1,7,42\}} \max(B_i, U_{is})$$

Two candidates in every oracle pair; reference labels choose the correct answer.

Decision check

Direct accuracy, rescue / harm,
and executable answer selection

Completed-grid sensitivity

All eight non-selected pair oracles;
baseline + nine crops for union coverage